

# Primal-dual proximal bundle and conditional gradient methods for convex problems

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**Objectives:** to develop primal-dual algorithmic frameworks, i.e., the inexact proximal point framework, that systematically guides the design and analysis of algorithms for convex problems such as convex optimization and convex-concave saddle-point problems.

- Primal-dual convergence of the proximal bundle (PB) method for solving convex and nonsmooth optimization;
- Primal-dual interpretation of conditional gradient and the cutting-plane scheme used within PB;
- Optimal PB-type method for convex-concave nonsmooth saddle-point problems.

# Review of Proximal Bundle Methods

The problem of interest is

$$\phi_* := \min_{x \in \mathbb{R}^n} \{\phi(x) := f(x) + h(x)\},$$

where  $f$  is convex and Lipschitz continuous, and  $h$  is a convex with a simple proximal mapping.

Approximately solve the proximal problem

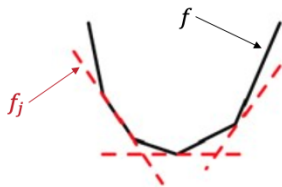
$$\hat{x} := \operatorname{argmin} \left\{ f(x) + h(x) + \frac{1}{2\lambda} \|x - x^c\|^2 \right\}$$

by an iterative process

$$x_j \leftarrow \min \left\{ f_j(x) + h(x) + \frac{1}{2\lambda} \|x - x^c\|^2 \right\}.$$

Recursively build up a cutting-plane model

$$f_j(x) = \max_{0 \leq i \leq j-1} \{\ell_f(x; x_i) := f(x_i) + \langle f'(x_i), x - x_i \rangle\}$$



# Generic Bundle Management

- First complexity result  $\mathcal{O}(\bar{\varepsilon}^{-3})$  (Kiwiel, 2000)
- Complexity in the strongly convex case  $\mathcal{O}((\mu\bar{\varepsilon})^{-1})$  (Du & Ruszczyński, 2017)
- Optimal complexity  $\mathcal{O}(\bar{\varepsilon}^{-2})$  (Liang & Monteiro, 2021, Díaz & Grimmer, 2023)
- A unified analysis for various bundle management (Liang & Monteiro, 2024)

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**Algorithm** Generic Bundle Management, GBM( $\lambda, \tau_j, x_0, x_j, \Gamma_j$ )

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**Initialize:**  $(\lambda, \tau_j) \in \mathbb{R}_{++} \times [0, 1]$ ,  $(x_0, x_j) \in \mathbb{R}^n \times \mathbb{R}^n$ , and  $\Gamma_j \in \overline{\text{Conv}}(\mathbb{R}^n)$  satisfying  $\Gamma_j \leq f$

- find a bundle model  $\Gamma_{j+1} \in \overline{\text{Conv}}(\mathbb{R}^n)$  satisfying  $\Gamma_{j+1} \leq f$  and

$$\Gamma_{j+1}(\cdot) \geq \tau_j \bar{\Gamma}_j(\cdot) + (1 - \tau_j) \ell_f(\cdot; x_j),$$

where  $\bar{\Gamma}_j \in \overline{\text{Conv}}(\mathbb{R}^n)$  satisfies  $\bar{\Gamma}_j \leq f$  and

$$\bar{\Gamma}_j(x_j) = \Gamma_j(x_j), \quad x_j = \underset{u \in \mathbb{R}^n}{\operatorname{argmin}} \left\{ \bar{\Gamma}_j(u) + h(u) + \frac{1}{2\lambda} \|u - x_0\|^2 \right\}.$$

**Output:**  $\Gamma_{j+1}$ .

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# Three examples

(E1) **single cut update:**  $\Gamma^+ = \Gamma_\tau^+ := \tau\Gamma + (1 - \tau)\ell_f(\cdot; x)$ .

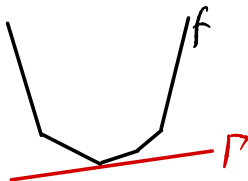
(E2) **two cuts update:** assume  $\Gamma = \max\{A_f, \ell_f(\cdot; x^-)\}$  where  $A_f$  is an affine function satisfying  $A_f \leq f$ , set

$$\Gamma^+ = \max\{A_f^+, \ell_f(\cdot; x)\}$$

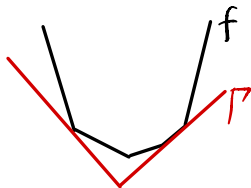
where  $A_f^+ = \theta A_f + (1 - \theta)\ell_f(\cdot; x^-)$ .

(E3) **multi cuts update:**  $\Gamma^+ = \max\{\Gamma, \ell_f(\cdot; x)\}$ .

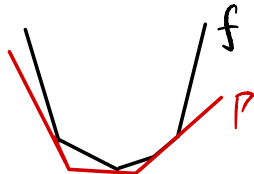
Bundle of past information  $\{(x_i, f(x_i), f'(x_i))\}$



Single cut



Two cuts



Multiple cuts

# Outline

- 1 Primal-dual Convergence
- 2 Duality between Cutting-plane and Conditional Gradient Methods
- 3 Saddle-point Problem
- 4 Numerical Experiments

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# Primal-dual Cutting-plane

The idea of the proximal bundle method is to recursively solve

$$\min_{u \in \mathbb{R}^n} \left\{ \phi^\lambda(u) := \phi(u) + \frac{1}{2\lambda} \|u - \hat{x}_{k-1}\|^2 \right\}$$

via the following recursive subproblem solver.

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**Algorithm** Primal-Dual Cutting-Plane, PDCP( $x_0, \lambda, \varepsilon$ )

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1. compute

$$x_j = \operatorname{argmin}_{u \in \mathbb{R}^n} \left\{ \Gamma_j(u) + h(u) + \frac{1}{2\lambda} \|u - x_0\|^2 \right\}, \quad s_j \in \partial \Gamma_j(x_j) \cap (-\partial h^\lambda(x_j)),$$

choose  $\tilde{x}_{j+1}$  such that

$$\phi^\lambda(\tilde{x}_{j+1}) \leq \tau_j \phi^\lambda(\tilde{x}_j) + (1 - \tau_j) \phi^\lambda(x_{j+1})$$

and set

$$m_j = \min_{u \in \mathbb{R}^n} \left\{ \Gamma_j(u) + h(u) + \frac{1}{2\lambda} \|u - x_0\|^2 \right\}, \quad t_j = \phi^\lambda(\tilde{x}_j) - m_j.$$

2. select  $\tau_j \in [0, 1]$  and update  $\Gamma_{j+1}$  by GBM( $\lambda, \tau_j, x_0, x_j, \Gamma_j$ ) and  $j \leftarrow j + 1$ ;

**Output:**  $(x_j, \tilde{x}_j, s_j)$  when  $t_j \leq \varepsilon$ .

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# Primal-dual convergence

We can show that  $t_j$  upper bounds the primal-dual gap of the proximal subproblem and converges at rate of  $\mathcal{O}(1/j)$

$$\phi^\lambda(\tilde{x}_j) + f^*(s_j) + (h^\lambda)^*(-s_j) \leq t_j \leq \frac{2t_1}{j(j+1)} + \frac{16\lambda M^2}{j+1}.$$

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**Algorithm** Primal-Dual Proximal Bundle, PDPB( $\hat{x}_0, \lambda, \bar{\varepsilon}$ )

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call oracle  $(\hat{x}_k, \tilde{x}_k, s_k) = \text{PDCP}(\hat{x}_{k-1}, \lambda, \bar{\varepsilon})$  and compute

$$\bar{x}_k = \frac{1}{k} \sum_{i=1}^k \tilde{x}_i, \quad \bar{s}_k = \frac{1}{k} \sum_{i=1}^k s_i. \quad (1)$$

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To find  $(\bar{x}_k, \bar{s}_k)$  such that the primal-dual gap is bounded by  $\bar{\varepsilon}$

$$\hat{\phi}(\bar{x}_k) + f^*(\bar{s}_k) + \hat{h}^*(-\bar{s}_k) \leq \bar{\varepsilon},$$

it takes  $\mathcal{O}(M^2 d_0^2 / \bar{\varepsilon}^2)$  iterations, where  $d_0 := \min_{x_* \in X_*} \{\|x_* - \hat{x}_0\|\}$ .

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# Dual viewpoint

Primal and dual proximal subproblems:  $f$  is convex and Lipschitz continuous, and  $h$  is convex

$$\min_{x \in \mathbb{R}^n} \left\{ f(x) + h(x) + \frac{1}{2\lambda} \|x - x_0\|^2 = f(x) + (h^\lambda)(x) \right\}, \quad (\text{P})$$

$$\min_{z \in \mathbb{R}^n} \left\{ (h^\lambda)^*(-z) + f^*(z) \right\}. \quad (\text{D})$$

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## Algorithm Regularized Cutting Plane for (P)

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$$x_j = \operatorname{argmin}_{x \in \mathbb{R}^n} \left\{ \Gamma_j(x) + (h^\lambda)(x) \right\}, \quad s_j \in \partial \Gamma_j(x_j) \cap (-\partial h^\lambda(x_j)),$$

$$\Gamma_{j+1}(\cdot) = \tau_j \Gamma_j(\cdot) + (1 - \tau_j) \ell_f(\cdot; x_j), \quad \tau_j = \frac{j}{j+2}.$$

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## Algorithm Conditional Gradient for (D)

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$$\bar{z}_j = \operatorname{argmin}_{z \in \mathbb{R}^n} \left\{ \langle -\nabla(h^\lambda)^*(-z_j), z \rangle + f^*(z) \right\},$$

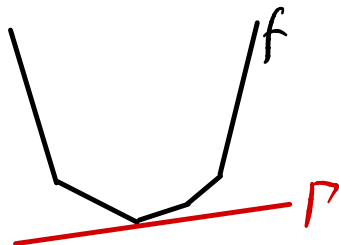
$$z_{j+1} = \tau_j z_j + (1 - \tau_j) \bar{z}_j.$$

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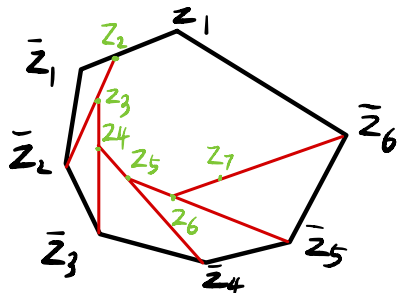
Equivalence  $s_j = z_j$ ,  $x_j = \nabla(h^\lambda)^*(-z_j)$ , and  $f'(x_j) = \bar{z}_j$ .

# Algorithmic duality visualized

Similar duality relationships have been discovered for CG and mirror descent (Bach, 2015), CG and dual averaging (Grigas, 2016), stochastic CG and randomized coordinate MD (Lu & Freund, 2021), and CG and PB (Fersztand & Sun, 2024).



Single cut



Standard CG

Equivalence  $s_j = z_j$ ,  $x_j = \nabla(h^\lambda)^*(-z_j)$ , and  $f'(x_j) = \bar{z}_j$ .

# GBM implementations inspired by CG

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## Algorithm Generic Bundle Management

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**Initialize:**  $(\lambda, \tau_j) \in \mathbb{R}_{++} \times [0, 1]$ ,  $(x_0, x_j) \in \mathbb{R}^n \times \mathbb{R}^n$ , and  $\Gamma_j \in \overline{\text{Conv}}(\mathbb{R}^n)$  satisfying  $\Gamma_j \leq f$

- find a bundle model  $\Gamma_{j+1} \in \overline{\text{Conv}}(\mathbb{R}^n)$  satisfying  $\Gamma_{j+1} \leq f$  and

$$\Gamma_{j+1}(\cdot) \geq \tau_j \bar{\Gamma}_j(\cdot) + (1 - \tau_j) \ell_f(\cdot; x_j),$$

where  $\bar{\Gamma}_j \in \overline{\text{Conv}}(\mathbb{R}^n)$  satisfies  $\bar{\Gamma}_j \leq f$  and

$$\bar{\Gamma}_j(x_j) = \Gamma_j(x_j), \quad x_j = \operatorname{argmin}_{u \in \mathbb{R}^n} \left\{ \bar{\Gamma}_j(u) + (h^\lambda)(u) \right\}.$$

**Output:**  $\Gamma_{j+1}$ .

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Previously choice  $\tau_j = \frac{j}{j+2}$ , other options inspired by CG

$$\alpha_j = \max \left\{ 0, 1 - \frac{S(z_j)}{\lambda \|z_j - \bar{z}_j\|^2} \right\},$$

$$\beta_j = \operatorname{argmin} \{ \psi(\beta z_j + (1 - \beta) \bar{z}_j) : \beta \in [0, 1] \}.$$

where  $\psi(z) = (h^\lambda)^*(-z) + f^*(z)$  is the dual function and  $S(z_j) = \phi^\lambda(x_j) + \psi(z_j)$  is the Wolfe gap.

# New variants of CG inspired by GBM implementations

General CG:  $z_j \in \text{Conv}\{z_1, \bar{z}_1, \dots, \bar{z}_{j-1}\} = \text{Conv}\{f'(x_0), f'(x_1), \dots, f'(x_{j-1})\}$

Two-cuts scheme for (P)

$$\Gamma_j(\cdot) = \max\{\bar{\Gamma}_{j-1}(\cdot), \ell_f(\cdot; x_{j-1})\}$$

$$\bar{\Gamma}_j(\cdot) = \theta_{j-1} \bar{\Gamma}_{j-1}(\cdot) + (1 - \theta_{j-1}) \ell_f(\cdot; x_{j-1})$$

$$\min_{(u,r) \in \mathbb{R}^n \times \mathbb{R}} \left\{ r + h^\lambda(u) : \bar{\Gamma}_{j-1}(u) \leq r, \ell_f(u, x_{j-1}) \leq r \right\}$$

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**Algorithm** Two-cuts scheme-inspired CG for (D)

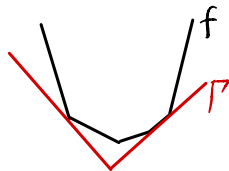
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$$z_j = \theta_{j-1} z_{j-1} + (1 - \theta_{j-1}) \bar{z}_{j-1} = \nabla \bar{\Gamma}_j,$$

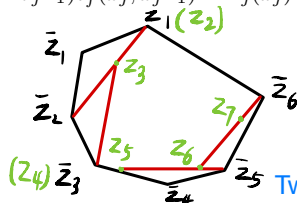
$$\bar{z}_j = \operatorname{argmin}_{z \in \mathbb{R}^n} \left\{ \langle -\nabla(h^\lambda)^*(-z_j), z \rangle + f^*(z) \right\} = f'(x_j).$$


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**Solve:**  $x_j = \nabla(h^\lambda)^*(-z_j)$  and  $\theta_{j-1} \bar{\Gamma}_{j-1}(x_j) + (1 - \theta_{j-1}) \ell_f(x_j; x_{j-1}) = \Gamma_j(x_j)$



Two cuts



Two-cuts CG

# New variants of CG inspired by GBM implementations

Multiple-cuts scheme for (P)

$$\Gamma_j(\cdot) = \max \{ \ell_f(\cdot; x_i) : i \in I_j \}$$

$$I_{j+1} = \bar{I}_{j+1} \cup \{j\}, \quad \bar{I}_{j+1} = \{i \in I_j : \theta_j^i > 0\}$$

$$\min_{(u,r) \in \mathbb{R}^n \times \mathbb{R}} \left\{ r + h^\lambda(u) : \ell_f(u; x_i) \leq r, \forall i \in I_j \right\}$$

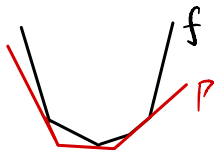
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**Algorithm** Multi-cuts scheme-inspired CG for (D)

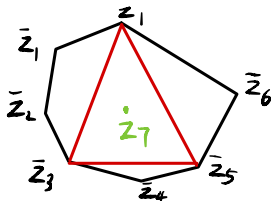
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$$z_j = \sum_{i \in I_j} \theta_j^i \bar{z}_i, \quad \bar{z}_j = f'(x_j)$$

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Multiple cuts



Multi-cuts CG

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# Saddle-point problem

## Convex-concave nonsmooth saddle-point problem

$$\min_{x \in \mathbb{R}^n} \max_{y \in \mathbb{R}^m} \{\phi(x, y) := f(x, y) + h_1(x) - h_2(y)\}$$

$\|f'_x(u, v)\| \leq M$ ,  $\|f'_y(u, v)\| \leq M$ , proximal mappings of  $h_1$  and  $h_2$  are simple,  $\text{dom } h_1 \times \text{dom } h_2$  is bounded with finite diameter  $D > 0$

A pair  $(x, y)$  is called a  $\bar{\varepsilon}$ -saddle-point if

$$\varphi(x) - \psi(y) \leq \bar{\varepsilon},$$

where

$$\varphi(u) = \max_{v \in \mathbb{R}^m} \phi(u, v), \quad \psi(v) = \min_{u \in \mathbb{R}^n} \phi(u, v).$$

## Composite subgradient method for SPP (CS-SPP)

$$\begin{aligned} x_k &= \operatorname{argmin}_{u \in \mathbb{R}^n} \left\{ \ell_{f(\cdot, y_{k-1})}(u; x_{k-1}) + h_1(u) + \frac{1}{2\lambda} \|u - x_{k-1}\|^2 \right\}, \\ y_k &= \operatorname{argmin}_{v \in \mathbb{R}^m} \left\{ -\ell_{f(x_{k-1}, \cdot)}(v; y_{k-1}) + h_2(v) + \frac{1}{2\lambda} \|v - y_{k-1}\|^2 \right\}. \end{aligned}$$

Letting  $\lambda = \bar{\varepsilon}/(32M^2)$ , then the iteration-complexity for CS-SPP to generate a  $\bar{\varepsilon}$ -saddle point is  $\mathcal{O}(M^2 D^2 / \bar{\varepsilon}^2)$ .

# Proximal bundle method

Proximal point formulation ( $\lambda_k = D/(4M\sqrt{k})$ )

$$(x_k, y_k) = \operatorname{argmin}_{x \in \mathbb{R}^n} \operatorname{argmax}_{y \in \mathbb{R}^m} \left\{ \phi(x, y) + \frac{1}{2\lambda_k} \|x - x_{k-1}\|^2 - \frac{1}{2\lambda_k} \|y - y_{k-1}\|^2 \right\}$$

Apply the cutting plane method (PDCP) to

$$\begin{aligned} \min_{x \in \mathbb{R}^n} & \left\{ f(x, y_{k-1}) + h_1(x) + \frac{1}{2\lambda_k} \|x - x_{k-1}\|^2 \right\}, \\ \min_{y \in \mathbb{R}^m} & \left\{ -f(x_{k-1}, y) + h_2(y) + \frac{1}{2\lambda_k} \|y - y_{k-1}\|^2 \right\}. \end{aligned}$$

Iteration-complexity  $\mathcal{O}((MD/\bar{\varepsilon})^{2.5})$  to find an  $\bar{\varepsilon}$ -saddle-point, for GBM.

Improved complexity  $\mathcal{O}((MD/\bar{\varepsilon})^2)$ , for two-cuts and multi-cuts schemes.

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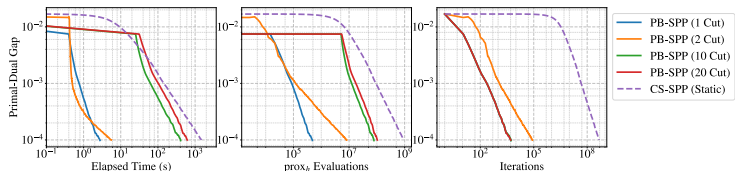
# Regularized matrix game

Consider a matrix game

$$\min_{x \in \Delta_n} \max_{y \in \Delta_m} \{y^\top Ax + \gamma_x \|x\|_\infty - \gamma_y \|y\|_\infty\}$$

where  $A \in \mathbb{R}^{m \times n}$  is the payoff matrix,  $x$  and  $y$  are mixed strategies on unit simplices. The  $\ell_\infty$  regularization terms with parameters  $\gamma_x \geq 0$  and  $\gamma_y \geq 0$  discourage overly concentrated strategies by penalizing large coordinates.

We compare five methods: CS-SPP with a static stepsize of  $\bar{\varepsilon}/(32M^2)$ , and PB-SPP with a dynamic stepsize of  $1/(2M\sqrt{k})$  under one-cut, two-cuts, 10-cuts, and 20-cuts schemes.



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Thank you!